

Econ 101A

Section 1

Anson Tam and Bruno Smaniotto*

January 22, 2025

1 Logical Languages¹

Mathematics is fundamentally a set of “languages” that we use to make logical statements, so a vital skill is being able to translate between these different languages. Here are the same three statements expressed in different languages.

- The language of necessary and sufficient conditions.
 1. B is a necessary condition for A .
 2. A is a sufficient condition for B .
 3. B and C together are necessary *and* sufficient conditions for A .
 4. It can also be the case that B is a necessary and sufficient condition for A , in which case they are said to be equivalent.
- The language of “if” statements.
 1. A only if B .
 2. B if A .
 3. A if and only if B and C (sometimes we write “ A iff B and C ”).
- The language of arrows.
 1. $A \Rightarrow B$
 2. $A \Rightarrow B$
 3. $A \Leftrightarrow B \text{ and } C$

*These notes are a consolidation of the notes of Jianmeng Lyu, Dan Acland, Ivan Balbuzanov, Mariana Carrera, Justin Gallagher, Anne Karing, Matt Leister, Sam Leone, Nicholas Li, Pablo Munoz, Yassine Sbair Sassi, Jon Schellenberg, Katalin Springel, Jonas Tungodden, David Wu, Sam Wang, and Matteo Saccarola.

¹For a reference, the greatest website in the world: https://en.wikipedia.org/wiki/List_of_logic_symbols.

2 Preferences and Utility Functions

2.1 Properties of Preferences

- $\succsim$ is **complete** if for all x and y in X either $x \succsim y$, or $y \succsim x$ or both.
- $\succsim$ is **transitive** if for all x, y , and z , $x \succsim y$ and $y \succsim z$ implies $x \succsim z$.
- $\succsim$ is **rational** if $\succsim$ is *complete* and *transitive*.
- A rational preference $\succsim$ is **continuous** if for all $x, y \in X$ such that $x \succ y$, we can find some $\epsilon > 0$ such that for all x' and y' that are less than ϵ distant from x and y , respectively, we have $x' \succ y'$.
- $\succsim$ is **reflexive** if $x \succsim x$ for every $x \in X$. (there's always a weak preference between a choice and itself).
- $\succsim$ is **monotonic** if $x \geq y$ implies $x \succsim y$ ²
- $\succsim$ is **strictly monotonic** if $x \geq y$ and $x_j > y_j$ for some j implies $x \succ y$.
- $\succsim$ is **convex** if for all x, y , and z in X such that $x \succsim z$ and $y \succsim z$, then $tx + (1 - t)y \succsim z$ for all t in $[0, 1]$.

2.2 Utility Function

Definition: A utility function $u : X \rightarrow \mathbb{R}$ represents $\succsim$ if, for every pair of points x and y in X , $x \succsim y$ if and only if $u(x) \geq u(y)$. In other words, if x has higher utility than y , you prefer x to y .

- **Theorem (Existence):** If preference relation $\succsim$ is rational and continuous on set X , there exists a continuous utility function $u : X \rightarrow \mathbb{R}$ that represents it.
- **Non-uniqueness:** The utility function representing preferences is not unique. If $u(x)$ represents preferences $\succsim$ and f is a strictly increasing function, then $f(u(x))$ represents $\succsim$ as well. For most of the time we only care about the **ordinal** property of utility function.

²If $x = (x_1, \dots, x_n)$ and $y = (y_1, \dots, y_n)$ are n -dimensional vectors, $x \geq y$ means that $x_j \geq y_j$ for $j = 1, \dots, n$

3 Implicit Function Theorem (IFT)

3.1 What is it for?

Doing comparative statics when you can't actually solve for optimal values (e.g. when you want to know how demand changes with price but you can't solve your first-order conditions for demand).

3.2 Why does it work?

For intuition: implicitly differentiate univariate function $f(x; p) \equiv 0$ with respect to p . Remember that x is a (univariate) function of p :

$$\frac{\partial f}{\partial p} + \frac{\partial f}{\partial x} \cdot \frac{dx}{dp} = 0$$

Rearrange to get result:

$$\frac{dx}{dp} = -\frac{\frac{\partial f}{\partial p}}{\frac{\partial f}{\partial x}}$$

Move p a little bit. If you know how much f changed as a direct result of the change in p and you also know how much f changed as an indirect result of the change in x (because p also changed x which in turn changed f), then you can "back out" how much p must have changed x given that $f(x; p)$ is always 0.

3.3 Univariate Version

Consider a function $f(x; p) = 0$ and a point (x_0, p_0) that solves this implicit function. In the most common case for this class: f is your F.O.C. from a maximization problem, p is a parameter like price, and x_0 is x^* (or a solution to the F.O.C).

- If the following 2 conditions hold:
 1. f continuously differentiable in a neighborhood of (x_0, p_0) (which is a solution to $f(x; p) = 0$).
 2. $f_x(x_0; p_0) \neq 0$
- Then we have 2 useful results:
 1. There is exactly 1 (univariate) function $g(p)$ such that $\forall p$ in a neighborhood of p_0 satisfy $f(g(p); p) = 0$ and $g(p_0) = x_0$.
 2. The derivative of $g(p)$ is

$$g'(p) = -\frac{f_p(g(p), p)}{f_x(g(p), p)} \quad (*)$$

Notice that since we defined g as a function that outputs x given an input value of p . $g'(p)$ is equivalent to $\frac{dx}{dp}$, answering the question "how does x change in response to a change in p , in order to keep the implicit function ($f(x; p) \equiv 0$) true?"

4 Exercises

4.1 Preferences and Utility Functions

Suppose there are two goods in the world: apples (x) and oranges (y), and that our preferences can be represented by the following utility function: $u(x, y) = 2x + y$

1. If a consumer does not have any apples, how many oranges would he need to be indifferent to a consumption bundle containing 2 apples and 1 orange?
2. Draw the indifference curves for when the utility equals 5 units and when the utility equals 7 units. Do these indifference curves cross? Can they ever cross? Why or why not?
3. Prove that the preferences represented by this utility function are transitive. That is, show that if $(x_1, y_1) \succsim (x_2, y_2)$ and $(x_2, y_2) \succsim (x_3, y_3)$, then $(x_1, y_1) \succsim (x_3, y_3)$.

4.2 Simple IFT

The function $f(x; p) = 1 - xp - p \cdot \exp(x) = 0$, around $(x_0, p_0) = (0, 1)$. Use the implicit function theorem to find $\frac{dx}{dp}$.